

Database Normalization Analysis Report

Disaster Management System

Executive Summary

This report presents a comprehensive normalization analysis of the Disaster Management System database. The original schema consisted of 8 tables, which have been analyzed through all normal forms (1NF, 2NF, 3NF, and BCNF). The normalization process identified multiple violations and resulted in a final schema of 19 tables, all in Boyce-Codd Normal Form (BCNF).

Table 1: DISASTER

Original Schema

disaster(disaster_id, disaster_type, location, start_date, end_date, security_level)

Functional Dependencies

- $\text{disaster_id} \rightarrow \text{disaster_type}, \text{location}, \text{start_date}, \text{end_date}, \text{security_level}$

Normalization Analysis

First Normal Form (1NF)

- Status:** ✔ Compliant
- Analysis:** All attributes contain atomic values with no repeating groups or multi-valued attributes. Each attribute holds a single, indivisible value.

Second Normal Form (2NF)

- Status:** ✔ Compliant
- Analysis:** The table has a single-attribute primary key (disaster_id). Therefore, no partial dependencies can exist. All non-key attributes are fully functionally dependent on the entire primary key.

Third Normal Form (3NF)

- Status:** ✔ Compliant
- Analysis:** No transitive dependencies exist. All non-key attributes depend directly on the primary key without any intermediate dependencies.

Boyce-Codd Normal Form (BCNF)

- **Status:**  Compliant
- **Analysis:** Every determinant is a candidate key. The only functional dependency has disaster_id as the determinant, which is the primary key.

Final Schema

disaster(disaster_id PK, disaster_type, location, start_date, end_date, security_level)

Outcome: No changes required. The table is already fully normalized.

Table 2: VICTIM

Original Schema

victim(victim_id, name, age, contact, disaster_id, shelter_id, shelter_address, disaster_type)

Functional Dependencies

- $\text{victim_id} \rightarrow \text{name, age, contact, disaster_id, shelter_id}$
- $\text{shelter_id} \rightarrow \text{shelter_address}$
- $\text{disaster_id} \rightarrow \text{disaster_type}$

Normalization Analysis

First Normal Form (1NF)

- **Status:**  Compliant
- **Analysis:** All attributes are atomic. Each column contains a single value per row.

Second Normal Form (2NF)

- **Status:**  Compliant
- **Analysis:** With victim_id as a single-attribute primary key, no partial dependencies can exist. All attributes depend on the complete primary key.

Third Normal Form (3NF)

- **Status:**  Not Compliant
- **Violations Identified:**
 1. **Transitive Dependency 1:** $\text{victim_id} \rightarrow \text{shelter_id} \rightarrow \text{shelter_address}$
 - The shelter_address depends on shelter_id, not directly on victim_id

2. Transitive Dependency 2: $\text{victim_id} \rightarrow \text{disaster_id} \rightarrow \text{disaster_type}$

- The disaster_type depends on disaster_id, not directly on victim_id

Normalization Steps: To achieve 3NF, we must eliminate transitive dependencies by creating separate tables for each transitive relationship.

Final Schema (BCNF)

```
victim(victim_id PK, name, age, contact, disaster_id FK, shelter_id FK)
```

```
victim_shelter_details(victim_id PK FK, shelter_address)
```

```
victim_disaster_details(victim_id PK FK, disaster_type)
```

Boyce-Codd Normal Form (BCNF)

- **Status:**  Compliant
- **Analysis:** All three tables now satisfy BCNF. Each determinant is a candidate key.

Tables Created: 2 additional tables

Table 3: SHELTER

Original Schema

```
shelter(shelter_id, address, capacity, coordinator_name, disaster_id)
```

Functional Dependencies

- $\text{shelter_id} \rightarrow \text{address, capacity, coordinator_name, disaster_id}$

Normalization Analysis

First Normal Form (1NF)

- **Status:**  Compliant
- **Analysis:** All attributes contain atomic values.

Second Normal Form (2NF)

- **Status:**  Compliant
- **Analysis:** Single-attribute primary key eliminates the possibility of partial dependencies.

Third Normal Form (3NF)

- **Status:**  Compliant
- **Analysis:** All non-key attributes depend directly on shelter_id with no transitive dependencies.

Boyce-Codd Normal Form (BCNF)

- **Status:**  Compliant
- **Analysis:** The only determinant (shelter_id) is the primary key.

Final Schema

shelter(shelter_id PK, address, capacity, coordinator_name, disaster_id FK)

Outcome: No changes required. The table is already fully normalized.

Table 4: VOLUNTEER

Original Schema

volunteer(volunteer_id, name, availability, volunteer_skills)

Functional Dependencies

- $\text{volunteer_id} \rightarrow \text{name, availability, volunteer_skills}$

Normalization Analysis

First Normal Form (1NF)

- **Status:**  Not Compliant
- **Violation:** The volunteer_skills attribute is multi-valued. A single volunteer can have multiple skills (e.g., "Medical, Communication, Logistics"), which violates the atomicity requirement of 1NF.

Normalization Steps: To achieve 1NF, we must separate the multi-valued attribute into a separate table with a composite primary key.

Normalized Schema

After 1NF:

volunteer(volunteer_id PK, name, availability)

volunteer_skills(volunteer_id FK, skill, PRIMARY KEY(volunteer_id, skill))

Second Normal Form (2NF)

- **Status:**  Compliant
- **Analysis:**
 - `volunteer` table: Single-attribute primary key prevents partial dependencies
 - `volunteer_skills` table: The composite key (`volunteer_id`, `skill`) fully determines all attributes (in this case, there are no non-key attributes)

Third Normal Form (3NF)

- **Status:**  Compliant
- **Analysis:** No transitive dependencies exist in either table.

Boyce-Codd Normal Form (BCNF)

- **Status:**  Compliant
- **Analysis:** All determinants are candidate keys.

Final Schema (BCNF)

volunteer(volunteer_id PK, name, availability)
volunteer_skills(volunteer_id FK, skill, PRIMARY KEY(volunteer_id, skill))

Tables Created: 1 additional table

Table 5: VOLUNTEER_ASSIGNMENTS

Original Schema

volunteer_assignments(volunteer_id, disaster_id, role, shift_timing, name, disaster_type)
Primary Key: (volunteer_id, disaster_id)

Functional Dependencies

- `(volunteer_id, disaster_id) → role, shift_timing, name, disaster_type`
- `volunteer_id → name`
- `disaster_id → disaster_type`

Normalization Analysis

First Normal Form (1NF)

- **Status:**  Compliant
- **Analysis:** All attributes are atomic.

Second Normal Form (2NF)

- **Status:**  Not Compliant
- **Violations Identified:**
 1. **Partial Dependency 1:** $\text{volunteer_id} \rightarrow \text{name}$
 - The name attribute depends on only part of the composite key (volunteer_id), not the entire key
 2. **Partial Dependency 2:** $\text{disaster_id} \rightarrow \text{disaster_type}$
 - The disaster_type depends on only disaster_id, not the full composite key

Normalization Steps: To achieve 2NF, we must remove attributes that depend on only part of the primary key.

Normalized Schema

After 2NF:

```
volunteer_assignments(volunteer_id FK, disaster_id FK, role, shift_timing,  
    PRIMARY KEY(volunteer_id, disaster_id))  
  
volunteer_assignment_names(volunteer_id PK FK, name)  
  
volunteer_assignment_disaster_types(disaster_id PK FK, disaster_type)
```

Third Normal Form (3NF)

- **Status:**  Compliant
- **Analysis:** After removing partial dependencies, no transitive dependencies remain.

Boyce-Codd Normal Form (BCNF)

- **Status:**  Compliant
- **Analysis:** All determinants in all three tables are candidate keys.

Final Schema (BCNF)

```
volunteer_assignments(volunteer_id FK, disaster_id FK, role, shift_timing,  
    PRIMARY KEY(volunteer_id, disaster_id))  
volunteer_assignment_names(volunteer_id PK FK, name)  
volunteer_assignment_disaster_types(disaster_id PK FK, disaster_type)
```

Tables Created: 2 additional tables

Table 6: RESOURCE

Original Schema

resource(resource_id, resource_name, description, disaster_id, shelter_id, quantity)

Functional Dependencies

- $\text{resource_id} \rightarrow \text{resource_name}, \text{description}$
- $(\text{resource_id}, \text{disaster_id}, \text{shelter_id}) \rightarrow \text{quantity}$

Primary Key Determination

The schema has an ambiguity regarding the primary key. Two scenarios exist:

Scenario 1: If $\text{PK} = \text{resource_id}$ alone

- This would mean a resource can only belong to one disaster and one shelter (unrealistic)

Scenario 2: If $\text{PK} = (\text{resource_id}, \text{disaster_id}, \text{shelter_id})$ (Assumed)

- This allows a resource to be distributed across multiple disasters and shelters with different quantities (realistic)

Normalization Analysis (Assuming Composite PK)

First Normal Form (1NF)

- Status:**  Compliant
- Analysis:** All attributes are atomic.

Second Normal Form (2NF)

- Status:**  Not Compliant
- Violation:** $\text{resource_id} \rightarrow \text{resource_name}, \text{description}$
 - These attributes depend only on resource_id, not on the complete composite key (resource_id, disaster_id, shelter_id)

Normalization Steps: Separate resource-specific information from location and quantity information.

Normalized Schema

After 2NF:

```
resource(resource_id PK, resource_name, description)
```

```
resource_location(resource_id FK, disaster_id FK, shelter_id FK, quantity,  
PRIMARY KEY(resource_id, disaster_id, shelter_id))
```

Third Normal Form (3NF)

- **Status:**  Compliant
- **Analysis:** No transitive dependencies exist.

Boyce-Codd Normal Form (BCNF)

- **Status:**  Compliant
- **Analysis:** All determinants are candidate keys.

Final Schema (BCNF)

```
resource(resource_id PK, resource_name, description)
```

```
resource_location(resource_id FK, disaster_id FK, shelter_id FK, quantity,  
PRIMARY KEY(resource_id, disaster_id, shelter_id))
```

Tables Created: 1 additional table

Table 7: INCIDENT_REPORT

Original Schema

```
incident_report(report_id, description, timestamp, volunteer_id, disaster_id,  
vol_name, disaster_type)
```

Functional Dependencies

- $\text{report_id} \rightarrow \text{description, timestamp, volunteer_id, disaster_id, vol_name, disaster_type}$
- $\text{volunteer_id} \rightarrow \text{vol_name}$
- $\text{disaster_id} \rightarrow \text{disaster_type}$

Normalization Analysis

First Normal Form (1NF)

- **Status:**  Compliant
- **Analysis:** All attributes contain atomic values.

Second Normal Form (2NF)

- **Status:**  Compliant
- **Analysis:** Single-attribute primary key (report_id) prevents partial dependencies.

Third Normal Form (3NF)

- **Status:**  Not Compliant
- **Violations Identified:**
 1. **Transitive Dependency 1:** $\text{report_id} \rightarrow \text{volunteer_id} \rightarrow \text{vol_name}$
 - vol_name depends on volunteer_id, which depends on report_id
 2. **Transitive Dependency 2:** $\text{report_id} \rightarrow \text{disaster_id} \rightarrow \text{disaster_type}$
 - disaster_type depends on disaster_id, which depends on report_id

Normalization Steps: Create separate tables for each transitive dependency.

Normalized Schema

After 3NF:

incident_report(report_id PK, description, timestamp, volunteer_id FK, disaster_id FK)

incident_volunteer_details(report_id PK FK, vol_name)

incident_disaster_details(report_id PK FK, disaster_type)

Boyce-Codd Normal Form (BCNF)

- **Status:**  Compliant
- **Analysis:** All three tables satisfy BCNF with all determinants being candidate keys.

Final Schema (BCNF)

incident_report(report_id PK, description, timestamp, volunteer_id FK, disaster_id FK)

incident_volunteer_details(report_id PK FK, vol_name)

incident_disaster_details(report_id PK FK, disaster_type)

Tables Created: 2 additional tables

Table 8: SHELTER_INVENTORY

Original Schema

shelter_inventory(shelter_id, resource_id, quantity, shelter_address, coord_name,
resource_name, disaster_id, disaster_type)
Primary Key: (shelter_id, resource_id)

Functional Dependencies

- (shelter_id, resource_id) → quantity, shelter_address, coord_name, resource_name, disaster_id, disaster_type
- shelter_id → shelter_address, coord_name, disaster_id
- resource_id → resource_name
- disaster_id → disaster_type

Normalization Analysis

First Normal Form (1NF)

- Status:

✔

 Compliant
- Analysis: All attributes are atomic.

Second Normal Form (2NF)

- Status:

✖

 Not Compliant
- Violations Identified:
 - Partial Dependency 1:

shelter_id → shelter_address, coord_name, disaster_id

 - These attributes depend only on shelter_id, not the complete composite key
 - Partial Dependency 2:

resource_id → resource_name

 - This attribute depends only on resource_id, not the complete composite key

Normalization Steps: Remove attributes that depend on only part of the primary key.

Normalized Schema

After 2NF:

```
shelter_inventory(shelter_id FK, resource_id FK, quantity,  
PRIMARY KEY(shelter_id, resource_id))
```

```
shelter_inventory_shelter_details(shelter_id PK FK, shelter_address, coord_name,  
disaster_id FK)
```

```
shelter_inventory_resource_details(resource_id PK FK, resource_name)
```

Third Normal Form (3NF)

- **Status:**  Not Compliant
- **Violation:** In the `shelter_inventory_shelter_details` table:
 - `shelter_id → disaster_id → disaster_type` (transitive dependency)
 - disaster_type depends on disaster_id, not directly on shelter_id

Normalization Steps: Create an additional table to eliminate the transitive dependency.

After 3NF:

```
shelter_inventory(shelter_id FK, resource_id FK, quantity,  
PRIMARY KEY(shelter_id, resource_id))
```

```
shelter_inventory_shelter_details(shelter_id PK FK, shelter_address, coord_name,  
disaster_id FK)
```

```
shelter_inventory_resource_details(resource_id PK FK, resource_name)
```

```
shelter_inventory_disaster_details(shelter_id FK, disaster_id PK FK, disaster_type)
```

Boyce-Codd Normal Form (BCNF)

- **Status:**  Compliant
- **Analysis:** All four tables now satisfy BCNF with all determinants being candidate keys.

Final Schema (BCNF)

```
shelter_inventory(shelter_id FK, resource_id FK, quantity,  
PRIMARY KEY(shelter_id, resource_id))
```

```
shelter_inventory_shelter_details(shelter_id PK FK, shelter_address, coord_name,  
disaster_id FK)
```

```
shelter_inventory_resource_details(resource_id PK FK, resource_name)
```

```
shelter_inventory_disaster_details(shelter_id FK, disaster_id PK FK, disaster_type)
```

Tables Created: 3 additional tables

Complete Final Normalized Database Schema (BCNF)

Core Tables (Normalized from Original)

1. **disaster**

disaster(disaster_id PK, disaster_type, location, start_date, end_date, security_level)

2. **victim**

victim(victim_id PK, name, age, contact, disaster_id FK, shelter_id FK)

3. **shelter**

shelter(shelter_id PK, address, capacity, coordinator_name, disaster_id FK)

4. **volunteer**

volunteer(volunteer_id PK, name, availability)

5. **volunteer_assignments**

volunteer_assignments(volunteer_id FK, disaster_id FK, role, shift_timing,
PRIMARY KEY(volunteer_id, disaster_id))

6. **resource**

resource(resource_id PK, resource_name, description)

7. **incident_report**

incident_report(report_id PK, description, timestamp, volunteer_id FK, disaster_id FK)

8. **shelter_inventory**

shelter_inventory(shelter_id FK, resource_id FK, quantity,
PRIMARY KEY(shelter_id, resource_id))

Supporting Tables (Created During Normalization)

9. volunteer_skills

volunteer_skills(volunteer_id FK, skill, PRIMARY KEY(volunteer_id, skill))

10. volunteer_assignment_names

volunteer_assignment_names(volunteer_id PK FK, name)

11. volunteer_assignment_disaster_types

volunteer_assignment_disaster_types(disaster_id PK FK, disaster_type)

12. resource_location

resource_location(resource_id FK, disaster_id FK, shelter_id FK, quantity,
PRIMARY KEY(resource_id, disaster_id, shelter_id))

13. victim_shelter_details

victim_shelter_details(victim_id PK FK, shelter_address)

14. victim_disaster_details

victim_disaster_details(victim_id PK FK, disaster_type)

15. incident_volunteer_details

incident_volunteer_details(report_id PK FK, vol_name)

16. incident_disaster_details

incident_disaster_details(report_id PK FK, disaster_type)

17. shelter_inventory_shelter_details

shelter_inventory_shelter_details(shelter_id PK FK, shelter_address, coord_name,
disaster_id FK)

18. shelter_inventory_resource_details

shelter_inventory_resource_details(resource_id PK FK, resource_name)

19. shelter_inventory_disaster_details

shelter_inventory_disaster_details(shelter_id FK, disaster_id PK FK, disaster_type)

Summary Table

Original Table	1NF Issue	2NF Issue	3NF Issue	Tables Created
disaster	✔ None	✔ None	✔ None	0
victim	✔ None	✔ None	✘ Transitive dependencies	2
shelter	✔ None	✔ None	✔ None	0
volunteer	✘ Multi-valued attribute	-	-	1
volunteer_assignments	✔ None	✘ Partial dependencies	-	2
resource	✔ None	✘ Partial dependencies	-	1
incident_report	✔ None	✔ None	✘ Transitive dependencies	2
shelter_inventory	✔ None	✘ Partial dependencies	✘ Transitive dependencies	3

Total Tables: 19 (8 original normalized + 11 new supporting tables)

Key Findings and Benefits

Normalization Violations Identified

- 1NF Violations: 1 table (multi-valued attributes)
- 2NF Violations: 3 tables (partial dependencies)
- 3NF Violations: 3 tables (transitive dependencies)

Benefits of Normalized Schema

1. **Data Integrity:** Elimination of redundancy reduces update anomalies
2. **Consistency:** Each piece of information is stored in exactly one place
3. **Maintainability:** Changes to data structures are localized
4. **Query Optimization:** Proper indexing on smaller, focused tables improves performance
5. **Scalability:** Normalized structure supports future expansion

Data Preservation

All original data has been preserved across the normalized tables. No information was lost during the normalization process; it was merely reorganized into a more efficient structure.

Conclusion

The Disaster Management System database has been successfully normalized to Boyce-Codd Normal Form (BCNF). The normalization process addressed multiple violations across different normal forms, resulting in a robust, efficient, and maintainable database structure. The final schema consists of 19 tables, all satisfying BCNF requirements, ensuring optimal data integrity and minimal redundancy.

End of Report